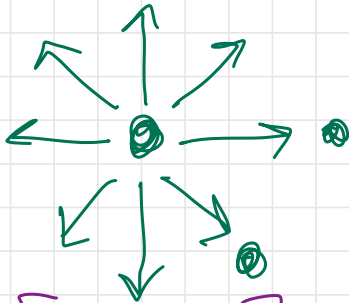


Backtracking

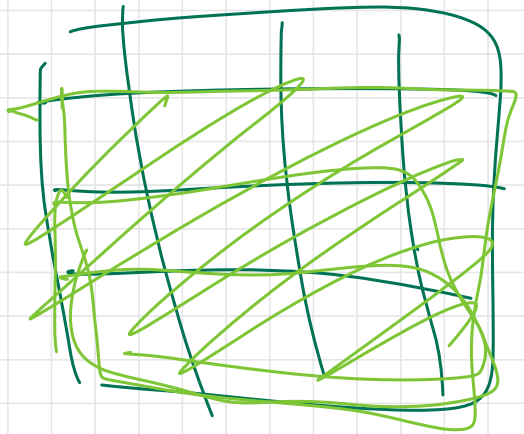
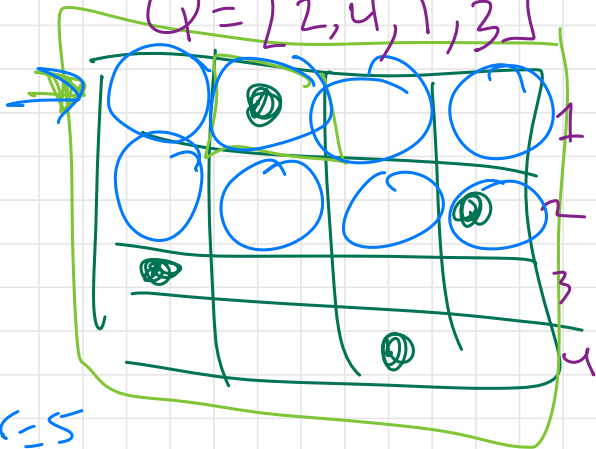
- ① Reduce any instance of a problem into slightly smaller instances of same problem
- ② try every choice

ex N-queens

place N queens on $N \times N$ chessboard
so that no pair can attack one another



$Q = [2, 4, 1, 3]$



choice: for each row, where to put its queen.

$Q[1..n]$ = placement of queen in i^{th} row
 r = rows already processed

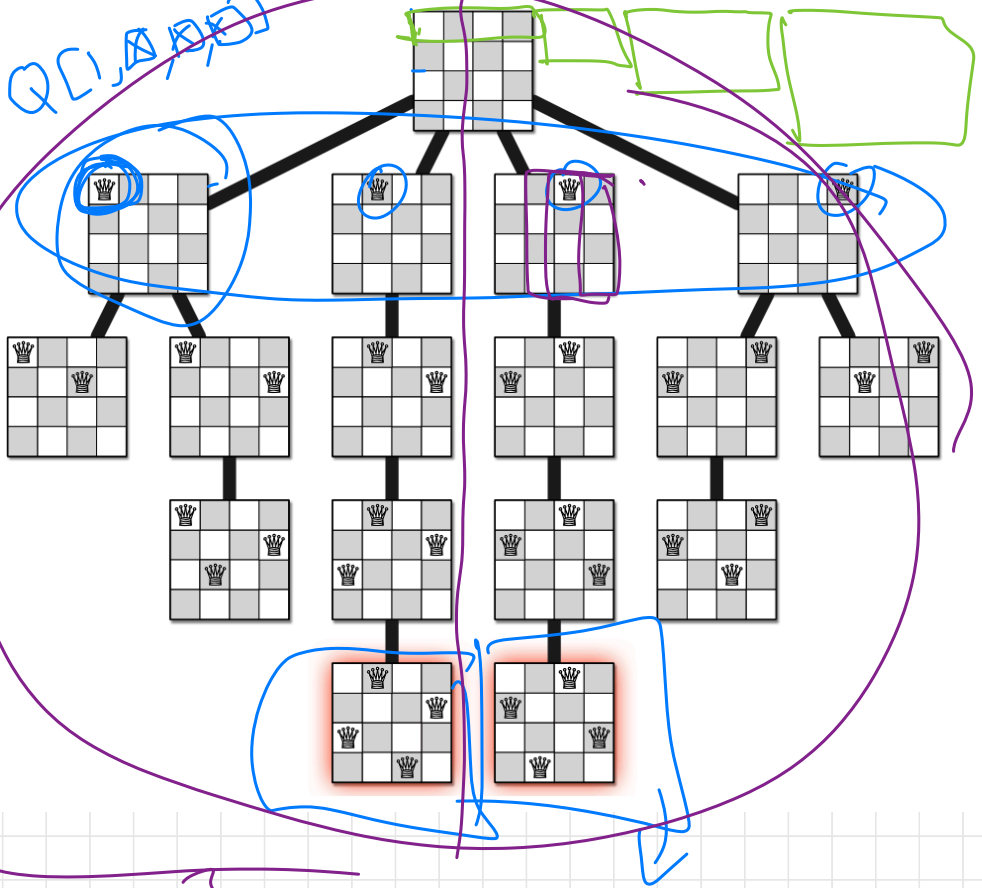
placeQueens(Q, r):

$\left[\begin{array}{l} r > n : \\ \quad \text{return } Q \end{array} \right.$

else:

$\left[\begin{array}{l} \text{for } j \text{ in } 1 \text{ to } n : \\ \quad \left[\begin{array}{l} \text{if placing queen at } j \text{ doesn't} \\ \text{attack any queens in } Q[1..r] : \\ \quad Q[r] = j \\ \quad \text{placeQueens}(Q, r+1) \end{array} \right. \end{array} \right.$

QCL, A, B, C



runtime ?

exponential

spitting text into words

PRIMVS DIGNITAS INTAM TENVISCI ENTIANONPOTEST
ESSERE SENIMSVNTPARVAE PROPEINSINGVLISLITTERIS
ATQVEINTERPVNCTIONIBUSVERBORVM OCCVPATAE

$T[1..n]$

Subroutine

isword(i, j): T if $T[i..j]$ is word
F otherwise

Is T splittable?

Choice: what is the next word?
is the suffix splittable

BLUE	STEM	UNIT	ROBOT	HEARTH	SATURNSPIN
------	------	------	-------	--------	------------

BLUE	STEM	UNIT	ROBOT	HE	ARTHANDSATURNSPIN
------	------	------	-------	----	-------------------

BLUE	STEM	UNIT	ROBOT	HEAR	THANDSATURNSPIN
------	------	------	-------	------	-----------------

BLUE	STEM	UNIT	ROBOT	HEART	HANDSATURNSPIN
------	------	------	-------	-------	----------------

BLUE	STEM	UNIT	ROBOT	HEARTH	ANDSATURNSPIN
------	------	------	-------	--------	---------------

BLUE STEM UNIT ROBOT HEA